

Intermittent Fasting Improves Insulin Sensitivity: A Randomised Trial

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Abstract

Intermittent fasting (IF) has emerged as a promising dietary strategy for improving metabolic health. This study investigates the impact of a 16:8 time-restricted eating (TRE) protocol on insulin sensitivity, fasting glucose, and HOMA-IR scores in overweight adults over a 12-week intervention.

Introduction

Insulin resistance is a hallmark of type 2 diabetes and metabolic syndrome. Dietary modifications that reduce postprandial glucose excursions have been shown to improve insulin signalling pathways. Intermittent fasting restricts the eating window, inducing metabolic switching from glucose to fatty-acid oxidation. Several animal models have demonstrated improved insulin sensitivity following IF protocols, yet human evidence remains heterogeneous. This study aims to fill this gap through a randomised controlled trial design.

Methods

One hundred adult volunteers (BMI 25-35) were randomised 1:1 to TRE (eating window 12:00-20:00) or ad libitum control for 12 weeks. Fasting venous blood samples were collected at baseline, week 6, and week 12. Insulin sensitivity was estimated using the Homeostatic Model Assessment for Insulin Resistance (HOMA-IR). Continuous glucose monitoring (CGM) devices provided 14-day glucose time-series at each visit. Statistical analysis used linear mixed-effects models with participant as random effect. The primary endpoint was change in HOMA-IR from baseline to week 12.

Results

The TRE group exhibited a mean HOMA-IR reduction of 1.8 units (95% CI 1.2-2.4, $p < 0.001$) compared to 0.3 units in controls ($p = 0.21$). Fasting insulin decreased by 22% in TRE versus 4% in controls. CGM-derived time in range improved from 71% to 83% in TRE participants. Body weight declined by 3.2 kg in TRE versus 0.6 kg in controls, though adjustment for weight loss did not attenuate the insulin-sensitivity effect ($p = 0.004$). Adverse events were mild and did not differ between arms.

Discussion

Our findings confirm that a 16:8 TRE protocol substantially improves insulin sensitivity independent of weight loss, consistent with mechanistic studies demonstrating circadian alignment of nutrient intake with peak insulin secretion. The magnitude of HOMA-IR improvement (1.8 units) exceeds that reported for metformin

monotherapy in early-stage insulin resistance. Limitations include the open-label design and reliance on self-reported eating windows. Future work should examine IF in combination with exercise and explore differential responses by chronotype.

Conclusion

Time-restricted eating for 12 weeks significantly reduces insulin resistance in overweight adults, with effects that persist after controlling for weight loss. These results support TRE as a scalable, low-cost metabolic intervention.

References

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